

Windows Authentication Monitoring & Threat Detection using Splunk

A Practical SIEM Project for Windows Security Log Analysis

Project Documentation

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Technologies Used

- Splunk Enterprise
 - Splunk Universal Forwarder
 - Windows Server
 - Windows Security Event Logs
 - VirtualBox
 - SPL (Search Processing Language)
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Project Objective

To design and implement a Windows Authentication Monitoring solution using Splunk that collects, analyzes, and visualizes Windows Security Event Logs. The project focuses on detecting successful logins, failed login attempts, privileged account usage, and authentication-related security events through dashboards and alerts.

Skills Demonstrated

- SIEM Implementation
- Splunk Administration

- Windows Log Analysis
 - SPL Query Development
 - Dashboard Creation
 - Security Monitoring
 - Threat Detection
 - Incident Investigation
 - Authentication Monitoring
 - Alert Configuration
-

Repository Structure

Windows-Authentication-Monitoring-Splunk/

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|  
|— Architecture/  
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|— Queries/  
|— Screenshots/  
|— README.md
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This document is intended for educational purposes and demonstrates a practical Security Information and Event Management (SIEM) implementation using Splunk. The project reflects hands-on experience in Windows authentication monitoring and SOC analyst workflows.

Acknowledgement

This project, **Windows Authentication Monitoring & Threat Detection using Splunk**, was developed to gain practical experience in Security Information and Event Management (SIEM) and Windows security monitoring. The primary objective of this project was to understand how security events are collected, analyzed, and visualized using Splunk Enterprise in a simulated Security Operations Center (SOC) environment.

Throughout this project, I explored the process of collecting Windows Security Event Logs using Splunk Universal Forwarder, writing Search Processing Language (SPL) queries, creating interactive dashboards, and configuring alerts to detect suspicious authentication activities. This hands-on implementation significantly improved my understanding of Windows authentication monitoring, log analysis, and security event investigation.

I would like to express my gratitude to the creators and maintainers of Splunk Enterprise for providing a powerful platform for log analysis and security monitoring. I also acknowledge Microsoft's official documentation for its detailed information on Windows Security Event Logs, which helped in understanding the significance of authentication-related events.

This project would not have been possible without continuous learning through official documentation, cybersecurity communities, online laboratories, and practical experimentation in a virtual environment.

Finally, I dedicate this project to my continuous learning journey in cybersecurity and my goal of becoming a skilled Security Operations Center (SOC) Analyst. I hope this documentation serves as a useful learning resource for students, beginners, and professionals who are interested in Windows security monitoring using Splunk.

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Chapter 1: Project Overview

1.1 Introduction

Security Operations Centers (SOCs) rely on Security Information and Event Management (SIEM) solutions to collect, analyze, and monitor security events generated across an organization's infrastructure. One of the most critical activities performed by a SOC is monitoring Windows authentication events to identify unauthorized access attempts, brute-force attacks, privileged account usage, and other suspicious login activities.

This project, **Windows Authentication Monitoring & Threat Detection using Splunk**, demonstrates the implementation of a practical SIEM solution using Splunk Enterprise. The project focuses on collecting Windows Security Event Logs, analyzing authentication events using Splunk Search Processing Language (SPL), creating dashboards for security monitoring, and configuring alerts to detect suspicious authentication behavior.

The project was developed in a virtual laboratory environment to simulate a real-world SOC workflow. Windows Security Event Logs were collected using Splunk Universal Forwarder and forwarded to Splunk Enterprise, where authentication events were indexed, analyzed, and visualized.

This documentation explains the complete implementation process, from environment setup and log collection to dashboard creation, alert configuration, testing, and troubleshooting.

1.2 Problem Statement

Windows operating systems continuously generate authentication events whenever users log in, log out, fail authentication, or perform privileged operations. In enterprise environments, thousands of these events are generated every day.

Manually reviewing Windows Event Viewer is not practical because:

- Large organizations generate thousands of authentication events every hour.
- Suspicious activities may be missed during manual analysis.
- Event correlation is difficult without centralized log management.
- Security investigations become time-consuming.

A centralized SIEM solution is therefore required to collect, analyze, and monitor authentication events efficiently.

1.3 Project Objectives

The primary objectives of this project are:

- Collect Windows Security Event Logs using Splunk Universal Forwarder.
 - Forward authentication events to Splunk Enterprise.
 - Monitor successful and failed login attempts.
 - Analyze Windows authentication events using SPL queries.
 - Create interactive dashboards for security monitoring.
 - Configure alerts for suspicious login activities.
 - Simulate a Security Operations Center (SOC) authentication monitoring workflow.
 - Develop practical experience with Splunk Enterprise and Windows Event Logs.
-

1.4 Scope of the Project

This project focuses specifically on Windows authentication monitoring. The implementation includes collecting and analyzing Windows Security Event Logs related to user authentication.

The project covers:

- Splunk Enterprise installation
- Splunk Universal Forwarder configuration
- Windows Security Event Log collection
- Authentication Event IDs (such as 4624 and 4625)
- SPL query development
- Dashboard creation
- Alert configuration
- Basic incident investigation
- Troubleshooting common issues

The project does not cover enterprise clustering, distributed Splunk deployments, or advanced threat hunting techniques.

1.5 Expected Outcome

After completing this project, the following outcomes were achieved:

- Windows authentication events were successfully collected in Splunk.
- Security logs were indexed and made searchable.
- SPL queries were developed to analyze authentication activity.
- Interactive dashboards were created to visualize login trends and user activity.
- Alerts were configured to detect repeated failed login attempts.
- A practical understanding of SIEM implementation and Windows authentication monitoring was achieved.

1.6 Technologies Used

Technology	Purpose
Splunk Enterprise	SIEM platform for log collection, indexing, searching, dashboards, and alerts
Splunk Universal Forwarder	Collects Windows Security Event Logs and forwards them to Splunk
Windows Operating System	Generates authentication-related Security Event Logs
Windows Event Viewer	Source of Security Event Logs
VirtualBox	Virtual lab environment
Search Processing Language (SPL)	Query language used for log analysis

1.7 Project Workflow

The implementation followed the workflow below:

1. Windows generated authentication events.
 2. Splunk Universal Forwarder collected Windows Security Event Logs.
 3. Logs were forwarded to Splunk Enterprise.
 4. Splunk indexed the incoming events.
 5. SPL queries analyzed authentication activities.
 6. Dashboards visualized authentication trends.
 7. Alerts notified the analyst when suspicious login activity was detected.
 8. Authentication events were investigated using Splunk Search & Reporting.
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Chapter Summary

This chapter introduced the Windows Authentication Monitoring project, explained the problem it addresses, defined the project objectives, described the technologies used, and outlined the complete workflow. The following chapter explains the laboratory environment used to implement the project before moving into the practical configuration of Splunk Enterprise.

Chapter 2: Lab Environment

2.1 Introduction

Before implementing the Windows Authentication Monitoring solution, a dedicated laboratory environment was created to simulate a basic Security Operations Center (SOC). The purpose of this lab was to safely collect, analyze, and monitor Windows Security Event Logs without affecting a production environment.

A virtualized environment was chosen because it provides flexibility, isolation, and the ability to test different security scenarios. Using virtualization also allowed the complete SIEM setup to be recreated whenever required.

The lab consisted of a Windows system generating authentication events and a Splunk Enterprise server responsible for collecting, indexing, searching, and visualizing those events.

2.2 Lab Objectives

The primary objectives of the laboratory environment were:

- Create a safe environment for learning SIEM technologies.
 - Collect Windows Security Event Logs.
 - Forward authentication logs to Splunk Enterprise.
 - Analyze authentication events using SPL queries.
 - Build security dashboards.
 - Configure alerts for suspicious login activities.
 - Simulate real-world SOC monitoring workflows.
-

2.3 Hardware Requirements

The project was implemented on a personal computer capable of running virtual machines.

Component	Minimum Requirement
Processor	Intel Core i5 / AMD Ryzen 5 or higher
RAM	8 GB (16 GB Recommended)
Storage	80 GB Free Disk Space

Operating System	Windows 10 / Windows 11 (64-bit)
Virtualization	Intel VT-x / AMD-V Enabled

These specifications were sufficient to run VirtualBox, Windows virtual machines, and Splunk Enterprise simultaneously.

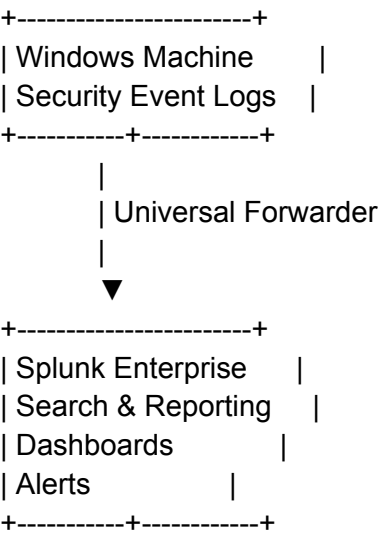
2.4 Software Requirements

The following software was used during the implementation of this project.

Software	Purpose
Oracle VirtualBox	Virtualization platform
Windows Operating System	Authentication event generation
Splunk Enterprise	SIEM platform
Splunk Universal Forwarder	Log forwarding
Windows Event Viewer	Security log generation and verification
Web Browser	Access Splunk Web Interface

2.5 Lab Topology

The project followed a simple architecture consisting of one Windows system and one Splunk Enterprise server.



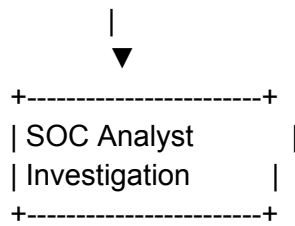


Figure 2.1 – Laboratory Environment

2.6 Network Configuration

Communication between the Windows machine and Splunk Enterprise was established using a virtual network.

The Windows machine forwarded Security Event Logs directly to the Splunk server through Splunk Universal Forwarder.

The communication process included:

1. Windows generated Security Events.
2. Universal Forwarder collected the events.
3. Events were forwarded to Splunk Enterprise.
4. Splunk indexed the incoming data.
5. SPL queries analyzed authentication events.
6. Dashboards displayed the collected information.

Proper network connectivity was verified before configuring log forwarding to ensure successful communication between both systems.

2.7 Why a Virtual Lab?

Using a virtual laboratory offers several advantages during cybersecurity learning and project development.

Some of the key benefits include:

- Safe testing environment.
- Easy recovery using snapshots.
- No impact on the host operating system.
- Cost-effective implementation.
- Ability to recreate attack scenarios.
- Practical experience with enterprise security tools.

For this project, virtualization provided an isolated environment where authentication events could be generated, collected, and analyzed without affecting production systems.

Chapter Summary

This chapter described the laboratory environment used throughout the project. It introduced the hardware and software requirements, explained the virtual network topology, and described how Windows Security Event Logs were forwarded to Splunk Enterprise. Establishing this environment provided the foundation for implementing the Windows Authentication Monitoring solution presented in the following chapters.

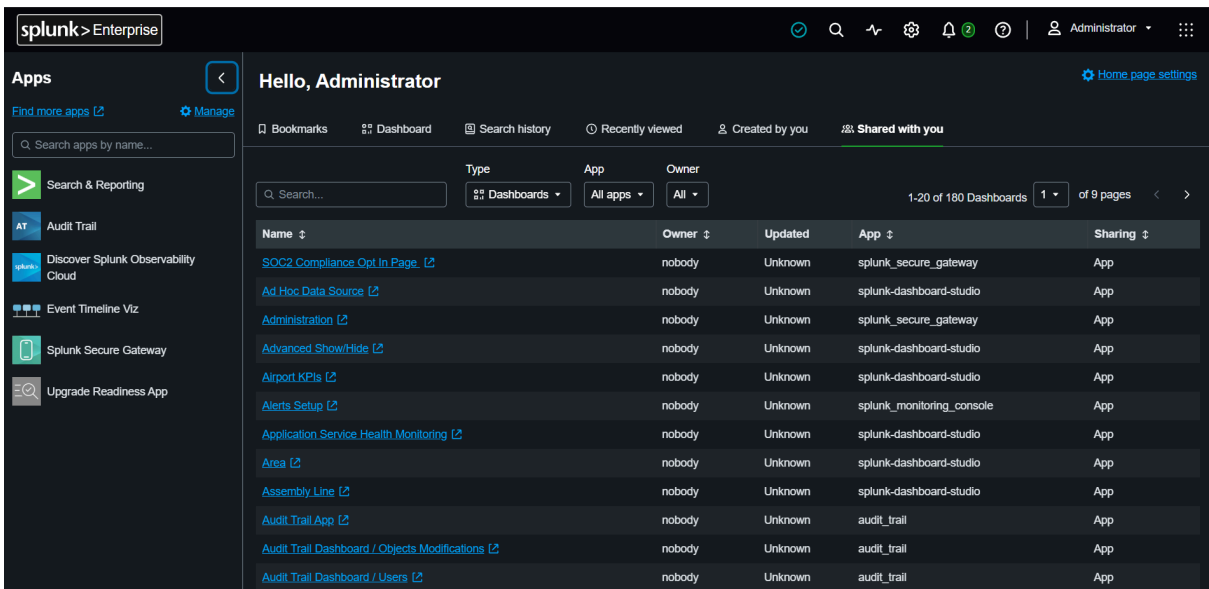
Chapter 4: Splunk Enterprise

4.1 Introduction to Splunk Enterprise

Splunk Enterprise is a Security Information and Event Management (SIEM) platform used to collect, index, search, analyze, and visualize machine-generated data from various sources. It enables organizations to monitor security events in real time, investigate incidents, and create alerts for suspicious activities.

In this project, Splunk Enterprise was used as the central monitoring platform for collecting Windows Security Event Logs. Authentication events generated by the Windows system were forwarded to Splunk using Splunk Universal Forwarder. After indexing the logs, SPL queries were executed to analyze login activity, create dashboards, and configure security alerts.

Figure 4.1 – Splunk Enterprise Home Interface



Description

Figure 4.1 shows the Splunk Enterprise interface used throughout this project. The **Search & Reporting** application was used to execute SPL queries, build dashboards, and investigate Windows authentication events.

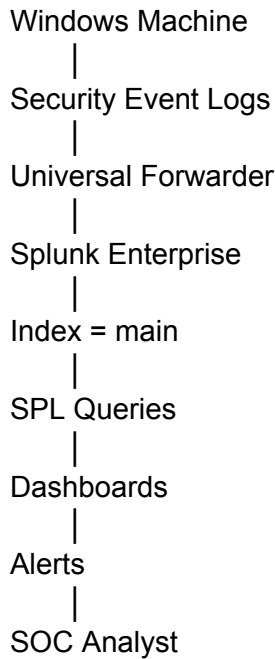
4.2 Role of Splunk in This Project

Splunk Enterprise acted as the central SIEM solution throughout the project.

Its responsibilities included:

- Receiving Windows Security Event Logs.
- Indexing authentication events.
- Executing SPL queries.
- Creating dashboards.
- Monitoring login activity.
- Generating security alerts.
- Supporting investigation of suspicious authentication events.

The complete workflow followed in this project is shown below.



4.3 Key Features Used

Although Splunk Enterprise provides hundreds of features, this project focused on the capabilities most relevant to a SOC analyst.

Feature	Purpose in This Project
Data Indexing	Store Windows Security Events
Search & Reporting	Execute SPL queries
Dashboards	Visualize authentication activity
Alerts	Detect brute-force attempts
Search Processing Language (SPL)	Analyze security logs

Event Search

Investigate authentication
events

These features enabled efficient monitoring and investigation of Windows authentication activity.

4.4 Data Processing Workflow

The following steps describe how Splunk processed authentication events during this project.

Step 1: Windows generated Security Event Logs.

Step 2: Splunk Universal Forwarder monitored the Security Log.

Step 3: Events were forwarded to Splunk Enterprise.

Step 4: Splunk indexed the received events into the **main** index.

Step 5: SPL queries searched and analyzed authentication events.

Step 6: Dashboards displayed authentication statistics.

Step 7: Alerts notified the SOC analyst whenever suspicious login activity was detected.

This workflow demonstrates the complete lifecycle of authentication monitoring using Splunk.

4.5 Splunk Components Used

The following Splunk components were implemented during the project.

Splunk Enterprise

Acts as the central SIEM platform responsible for indexing, searching, dashboard creation, and alert generation.

Universal Forwarder

Installed on the Windows machine to collect Security Event Logs and forward them to Splunk Enterprise.

Search & Reporting App

Used to execute SPL queries, investigate logs, and build dashboards.

Main Index

All Windows authentication events were stored in the **main** index, making them searchable using SPL.

4.6 Benefits Achieved

Using Splunk Enterprise provided several advantages during the implementation of this project.

- Centralized log collection.
- Faster authentication analysis.
- Real-time monitoring.
- Interactive dashboards.
- Automated alert generation.
- Simplified security investigations.
- Practical experience with enterprise SIEM workflows.

These capabilities demonstrate why Splunk is widely used within Security Operations Centers (SOCs).

4.7 Implementation Summary

The implementation of Splunk Enterprise in this project successfully demonstrated a complete Windows authentication monitoring workflow.

Authentication events generated by Windows were collected using Splunk Universal Forwarder, indexed into Splunk Enterprise, analyzed using SPL queries, and presented through dashboards and alerts. This implementation provided hands-on experience with log management, event analysis, and basic threat detection within a simulated SOC environment.

Chapter Summary

This chapter introduced Splunk Enterprise and explained its role in the Windows Authentication Monitoring project. It described how authentication events were collected, indexed, analyzed, and visualized using Splunk. The next chapter focuses on **Splunk Universal Forwarder**, including its installation, configuration, and role in forwarding Windows Security Event Logs to Splunk Enterprise.

Chapter 5: Splunk Universal Forwarder

5.1 Introduction

Splunk Universal Forwarder (UF) is a lightweight software component used to collect logs from source systems and forward them to Splunk Enterprise. It is designed to consume minimal system resources while continuously monitoring log files and Windows Event Logs.

In this project, the Universal Forwarder was installed on the Windows machine to collect authentication-related Security Event Logs and forward them to the Splunk Enterprise server for analysis.

5.2 Purpose of Universal Forwarder

The primary purpose of the Universal Forwarder is to automate log collection. Instead of manually exporting Windows Event Logs, the forwarder continuously monitors the configured log source and sends new events to Splunk Enterprise in near real time.

Using the Universal Forwarder provided the following benefits:

- Automated log collection.
 - Continuous event forwarding.
 - Low CPU and memory usage.
 - Reliable communication with Splunk Enterprise.
 - Real-time security monitoring.
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5.3 Role in This Project

Within this project, the Universal Forwarder acted as the bridge between the Windows operating system and Splunk Enterprise.

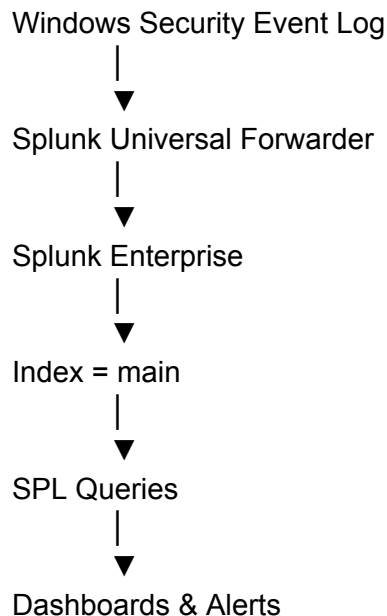
The workflow was as follows:

1. Windows generated Security Event Logs.
2. Universal Forwarder detected newly generated events.
3. Events were forwarded to Splunk Enterprise.
4. Splunk indexed the logs into the **main** index.
5. SPL queries analyzed the collected authentication events.
6. Dashboards and alerts displayed the results.

Without the Universal Forwarder, Windows authentication events would not be available for analysis in Splunk.

5.4 Log Forwarding Process

The following diagram illustrates how authentication events were forwarded during the project.



This workflow enabled continuous monitoring of authentication activity without manual intervention.

5.5 Advantages of Using Universal Forwarder

The Universal Forwarder provides several advantages in enterprise environments.

- Lightweight and efficient.
- Supports real-time log forwarding.
- Easy to configure.
- Secure communication with Splunk Enterprise.
- Scalable for thousands of endpoints.
- Minimal impact on system performance.

These features make it one of the most widely used log collection agents in enterprise SIEM deployments.

5.6 Implementation Summary

The Universal Forwarder was successfully configured to collect Windows Security Event Logs and send them to Splunk Enterprise. After forwarding the events, Splunk indexed the data and made it available for searching using SPL queries.

This configuration formed the foundation of the Windows Authentication Monitoring solution implemented in this project.

Chapter Summary

This chapter explained the purpose and role of Splunk Universal Forwarder in the Windows Authentication Monitoring project. It described how authentication events were collected from the Windows system and forwarded to Splunk Enterprise for indexing and analysis. The next chapter focuses on Windows Security Event Logs and explains the authentication events monitored during the project.

Chapter 6: Windows Security Event Logs

6.1 Introduction

Windows Security Event Logs are one of the most important sources of information for Security Operations Centers (SOCs). Every authentication attempt, account modification, privilege assignment, and security-related activity performed on a Windows system is recorded as an event.

These logs help security analysts monitor user activity, investigate incidents, and detect suspicious behavior such as brute-force attacks, unauthorized logins, and privilege escalation attempts.

In this project, Windows Security Event Logs were collected using Splunk Universal Forwarder and forwarded to Splunk Enterprise for monitoring and analysis.

6.2 What are Windows Security Event Logs?

Windows Security Event Logs are records generated by the Windows operating system whenever a security-related action occurs.

Each event contains detailed information such as:

- Event ID
- Date and Time
- Username
- Computer Name
- Logon Type
- Source IP Address (when available)
- Status of the operation

These details allow SOC analysts to understand what happened, when it happened, and who performed the activity.

6.3 Why These Logs Were Collected

The primary objective of this project was to monitor Windows authentication activities.

Collecting Windows Security Event Logs allowed the project to:

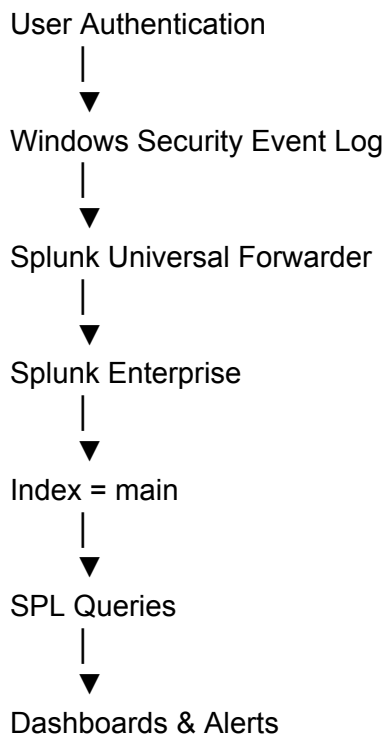
- Monitor successful logins.
- Detect failed login attempts.

- Identify privileged account logins.
- Analyze user authentication patterns.
- Create dashboards for visualization.
- Generate alerts for suspicious authentication activity.

These logs formed the primary data source used throughout the project.

6.4 Security Event Log Workflow

The following workflow describes how authentication events were processed.



This automated workflow enabled continuous monitoring of Windows authentication events.

6.5 Authentication Events Monitored

This project focused on monitoring the following Windows Security Event IDs.

Event ID	Description	Purpose
4624	Successful Login	Track successful user authentication

4625	Failed Login	Detect failed authentication attempts
4672	Special Privileges Assigned	Monitor privileged account logins
4740	Account Lockout	Detect locked user accounts

These Event IDs were selected because they are commonly monitored by SOC analysts to detect authentication-related threats.

Each Event ID will be explained in detail in the next chapter.

6.6 Importance in SOC Operations

Windows Security Event Logs provide valuable visibility into user authentication activities.

SOC analysts use these logs to:

- Investigate suspicious login attempts.
- Detect brute-force attacks.
- Identify privileged account usage.
- Review account lockout events.
- Correlate authentication activity with other security events.

Monitoring these events helps organizations detect security incidents at an early stage.

6.7 Implementation Summary

In this project, Windows Security Event Logs served as the primary source of authentication data.

The logs were collected from the Windows system, forwarded to Splunk Enterprise, indexed into the **main** index, and analyzed using SPL queries. The analyzed data was then displayed through dashboards and monitored using alerts.

This implementation demonstrates a practical approach to Windows authentication monitoring using Splunk.

Chapter Summary

This chapter introduced Windows Security Event Logs and explained their importance in authentication monitoring. It described the authentication events collected during the project and how they were processed by Splunk Enterprise. The next chapter explains each monitored Event ID in detail and discusses its significance in security monitoring.

Chapter 7: Windows Authentication Event IDs

7.1 Introduction

Windows Security Event Logs assign a unique Event ID to every security-related activity performed on the operating system. These Event IDs help administrators and SOC analysts quickly identify the type of event without reviewing the entire log.

In this project, four important authentication Event IDs were monitored using Splunk Enterprise. These events were selected because they provide valuable information about user authentication, privileged access, and possible security threats.

7.2 Event ID 4624 – Successful Logon

Event ID **4624** is generated whenever a user successfully authenticates to the Windows system.

This event confirms that the username and credentials were accepted and access was granted.

Why it is Important

Monitoring successful logins helps security teams:

- Track user activity.
- Verify legitimate access.
- Identify unusual login times.
- Detect unauthorized account usage.

How It Was Used in This Project

Event ID 4624 was used to create the **Top Successful Users** dashboard, allowing authentication activity to be summarized by user account.

This helped identify which users logged into the system most frequently.

7.3 Event ID 4625 – Failed Logon

Event ID **4625** is generated whenever a user fails to authenticate successfully.

Common reasons include:

- Incorrect password.
- Invalid username.
- Disabled account.
- Expired password.

Why it is Important

Repeated failed login attempts may indicate:

- Brute-force attacks.
- Password guessing.
- Unauthorized access attempts.

Monitoring this event enables SOC analysts to detect suspicious authentication behavior before an account is compromised.

How It Was Used in This Project

Event ID 4625 was used to create:

- Failed Login Trend Dashboard.
- Brute Force Detection Alert.

The alert notified the analyst whenever repeated failed login attempts exceeded the configured threshold.

7.4 Event ID 4672 – Special Privileges Assigned

Event ID **4672** is generated whenever a user logs in with administrative or other high-level privileges.

This event indicates that the account has elevated permissions capable of performing sensitive operations.

Why it is Important

Privileged accounts are attractive targets for attackers because they provide extensive access to systems and data.

Monitoring privileged logins helps identify:

- Administrator logins.
- Unexpected privileged access.
- Potential privilege abuse.

How It Was Used in This Project

This Event ID was used to create the **Privileged Logins** dashboard, allowing administrative account activity to be monitored separately from normal user accounts.

7.5 Event ID 4740 – Account Lockout

Event ID **4740** is generated whenever a Windows account becomes locked after multiple failed login attempts.

Why it is Important

Account lockouts often indicate:

- Password guessing attacks.
- Brute-force attacks.
- Users repeatedly entering incorrect passwords.

Monitoring this event helps security teams respond quickly to authentication-related security incidents.

How It Was Used in This Project

Although Event ID 4740 was included in the monitoring configuration, no account lockout events were generated during project testing. It remains an important Event ID for future expansion of the project.

7.6 Summary of Event IDs

Event ID	Description	Project Usage
4624	Successful Logon	Top Successful Users Dashboard
4625	Failed Logon	Failed Login Trend Dashboard and Brute Force Alert
4672	Special Privileges Assigned	Privileged Logins Dashboard
4740	Account Lockout	Future monitoring and incident detection

7.7 Implementation Summary

These authentication Event IDs formed the foundation of the Windows Authentication Monitoring solution. By collecting and analyzing these events in Splunk Enterprise, it was possible to monitor user authentication, detect failed logins, identify privileged account activity, and configure security alerts for suspicious behavior.

Chapter Summary

This chapter explained the Windows Security Event IDs monitored throughout the project. Each Event ID provided valuable authentication information that was used to build dashboards, create alerts, and support security investigations. In the next chapter, the SPL queries used to analyze these events will be explained in detail.

Chapter 8: Search Processing Language (SPL) Queries

8.1 Introduction

Search Processing Language (SPL) is Splunk's query language used to search, filter, analyze, and visualize machine data. It enables security analysts to investigate authentication events, identify suspicious activities, and generate meaningful insights from Windows Security Event Logs.

In this project, SPL queries were developed to monitor Windows authentication events such as successful logins, failed login attempts, privileged account usage, and logon type distribution. These queries formed the foundation of the dashboards and alerts created during the implementation.

8.2 Failed Login Trend

Objective

Monitor failed Windows login attempts over time to identify suspicious authentication activity.

SPL Query

```
index=main EventCode=4625  
| timechart count
```

Query Explanation

- **index=main** → Searches the main index where Windows Security Event Logs are stored.
- **EventCode=4625** → Filters only failed login events.
- **timechart count** → Displays the number of failed login attempts over time.

SOC Use Case

This query helps analysts detect:

- Brute-force attacks.
- Password guessing attempts.
- Sudden increases in failed authentication events.

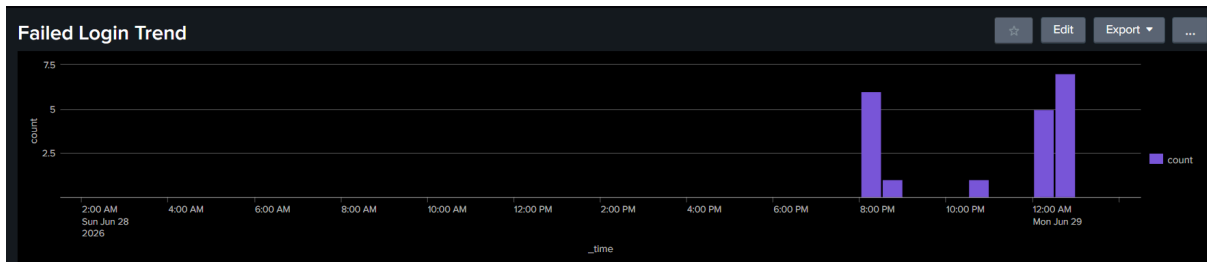


Figure 8.1 – Failed Login Trend Dashboard

8.3 Top Successful Users

Objective

Identify the users who logged in successfully most frequently.

SPL Query

```
index=main EventCode=4624  
| top Account_Name
```

Query Explanation

- **index=main** → Searches the Windows Security Event Logs.
- **EventCode=4624** → Filters successful login events.
- **top Account_Name** → Displays the users with the highest number of successful logins.

SOC Use Case

This query helps analysts:

- Monitor user login activity.
- Identify frequently used accounts.
- Detect unusual authentication behavior.

Top Successful Users	
Account_Name ↕	count ↕
WIN-VACB12M6K0S\$	986
-	949
SYSTEM	488
DANISH_MANSURI\$	223
DEMO-PCS	178
testuser1	47
danishmansuri488@gmail.com	24
Administrator	12
DWM-1	6
UMFD-0	5
UMFD-1	5
DWM-2	4
LOCAL_SERVICE	3
NETWORK_SERVICE	3
UMFD-2	3
DWM-3	2
DWM-64	2
DWM-65	2
DWM-66	2
danishm@gmail.com	2

Figure 8.2 – Top Successful Users Dashboard

8.4 Privileged Logins

Objective

Monitor logins performed using privileged or administrative accounts.

SPL Query

```
index=main EventCode=4672
| stats count by Account_Name
```

Query Explanation

- **EventCode=4672** → Searches privileged logon events.
- **stats count by Account_Name** → Counts privileged logins for each account.

SOC Use Case

This query enables analysts to:

- Monitor administrator account activity.
- Detect unexpected privileged access.
- Investigate privilege abuse.

Privileged Logins	
Account_Name ↕	count ↕
Administrator	16
DEMO-PC\$	4
DWM-1	6
DWM-2	4
DWM-3	2
DWM-64	2
DWM-65	2
DWM-66	2
LOCAL_SERVICE	3
NETWORK_SERVICE	3
SYSTEM	485
WIN-VACB12M6KOS\$	842
danishm@gmail.com	1
danishmansuri4888@gmail.com	12

Figure 8.3 – Privileged Logins Dashboard

8.5 Logon Type Distribution

Objective

Analyze the different logon types used during authentication.

SPL Query

```
index=main EventCode=4624
| chart count by Logon_Type
```

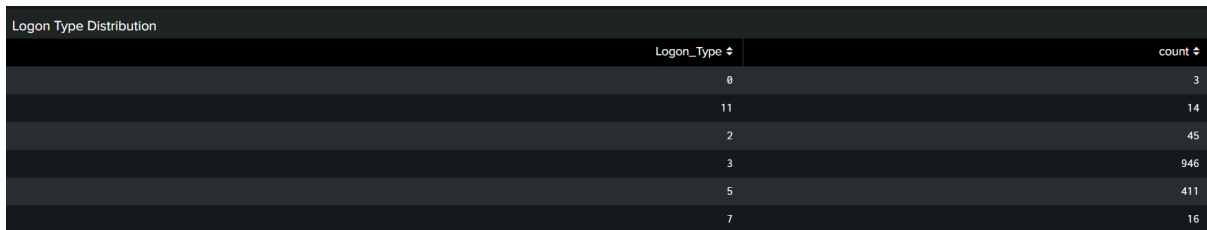
Query Explanation

- **EventCode=4624** → Searches successful login events.
- **chart count by Logon_Type** → Groups successful logins according to logon type.

SOC Use Case

This query helps identify:

- Interactive logins.
- Remote Desktop logins.
- Service account logins.
- Network logins.

A dashboard titled "Logon Type Distribution" showing a table of logon types and their counts. The table has two columns: "Logon_Type" and "count". The data is as follows:

Logon_Type	count
0	3
11	14
2	45
3	946
5	411
7	16

Figure 8.4 – Logon Type Distribution Dashboard

8.6 Summary

The SPL queries developed in this project transformed raw Windows Security Event Logs into meaningful security insights. By filtering authentication events, grouping related activities, and visualizing the results, these queries enabled continuous monitoring of Windows authentication activity. They also formed the foundation for the dashboards and security alerts implemented during the project.

Chapter Summary

This chapter explained the SPL queries used to analyze Windows authentication events in Splunk Enterprise. Each query served a specific monitoring objective and supported the creation of dashboards that improved visibility into user authentication activity. The next chapter explains how these dashboards were designed and how they assist SOC analysts during security investigations.

Chapter 9: Dashboard Design and Analysis

9.1 Introduction

Dashboards play a critical role in Security Information and Event Management (SIEM) by transforming raw security events into meaningful visual information. Instead of manually reviewing thousands of Windows Security Event Logs, dashboards enable Security Operations Center (SOC) analysts to quickly identify suspicious activities, monitor authentication trends, and investigate potential security incidents.

In this project, multiple dashboards were created using Splunk Enterprise to visualize Windows authentication events. Each dashboard focuses on a specific aspect of authentication monitoring and provides valuable insights for security investigations.

9.2 Failed Login Trend Dashboard

Objective

The Failed Login Trend dashboard displays failed authentication attempts over time using Windows Event ID **4625**.

By monitoring failed login trends, SOC analysts can quickly identify unusual spikes in authentication failures that may indicate brute-force attacks or password guessing attempts.

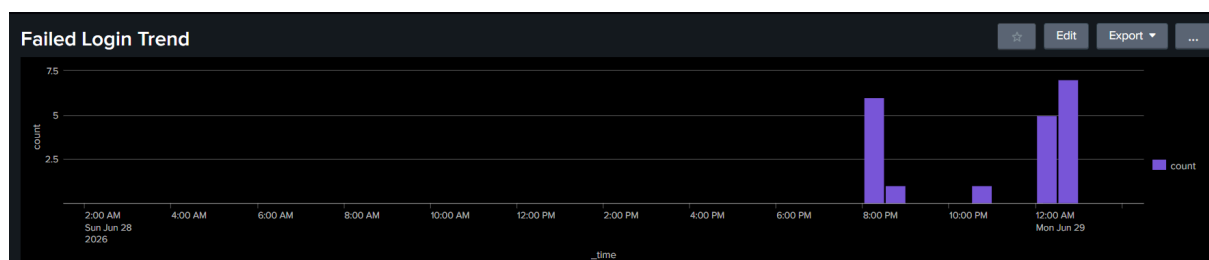


Figure 9.1 – Failed Login Trend Dashboard

SOC Analysis

During normal system activity, failed login attempts are usually low and consistent. However, a sudden increase in failed authentication events should be investigated immediately.

Possible causes include:

- Brute-force attacks
- Incorrect password attempts
- User account lockout attempts
- Automated login scripts

9.3 Top Successful Users Dashboard

Objective

This dashboard displays the users with the highest number of successful logins using Windows Event ID **4624**.

It helps security teams understand authentication patterns and identify accounts with unusually high login activity.

Top Successful Users	
Account_Name ↕	count ↕
WIN-VACB12M6K0SS	986
-	949
SYSTEM	488
DANISH_MANSURI\$	223
DEMO-PCS	178
testuser1	47
danishmansuri488@gmail.com	24
Administrator	12
DWM-1	6
UMFD-0	5
UMFD-1	5
DWM-2	4
LOCAL_SERVICE	3
NETWORK_SERVICE	3
UMFD-2	3
DWM-3	2
DWM-64	2
DWM-65	2
DWM-66	2
danishm@gmail.com	2

Figure 9.2 – Top Successful Users Dashboard

SOC Analysis

SOC analysts can use this dashboard to:

- Monitor frequently used accounts.
- Detect unexpected user activity.
- Identify service accounts with abnormal authentication behavior.
- Investigate unusual login frequency.

High authentication activity does not always indicate malicious behavior, but it should be reviewed if it differs from normal user behavior.

9.4 Privileged Logins Dashboard

Objective

The Privileged Logins dashboard monitors authentication events generated by privileged accounts using Windows Event ID **4672**.

Administrative accounts have elevated permissions and therefore require continuous monitoring.



The screenshot shows a dashboard titled 'Privileged Logins' with a table listing account names and their corresponding login counts. The table has two columns: 'Account_Name' and 'count'. The data is as follows:

Account_Name	count
Administrator	16
DEMO-PC\$	4
DWM-1	6
DWM-2	4
DWM-3	2
DWM-64	2
DWM-65	2
DWM-66	2
LOCAL SERVICE	3
NETWORK SERVICE	3
SYSTEM	405
WIN-VACB12M6KOS\$	842
danishm@gmail.com	1
danishmansuri4888@gmail.com	12

Figure 9.3 – Privileged Logins Dashboard

SOC Analysis

This dashboard helps analysts:

- Monitor administrator account usage.
- Detect unexpected privileged logins.
- Identify privilege escalation attempts.
- Verify administrative activities.

Because privileged accounts have extensive access to systems and data, any unusual authentication activity should be investigated promptly.

9.5 Logon Type Distribution Dashboard

Objective

This dashboard categorizes successful logins according to Windows Logon Type.

Different logon types represent different methods of authentication, such as interactive logins, network logins, service logins, or Remote Desktop sessions.

Privileged Logins	
Account_Name ↕	count ↕
Administrator	16
DEMO-PC\$	4
DWM-1	6
DWM-2	4
DWM-3	2
DWM-64	2
DWM-65	2
DWM-66	2
LOCAL_SERVICE	3
NETWORK_SERVICE	3
SYSTEM	485
WIN-VACB12MEKOS\$	842
danishm@gmail.com	1
danishmansuri4888@gmail.com	12

Figure 9.4 – Logon Type Distribution Dashboard

SOC Analysis

Monitoring logon types enables analysts to:

- Understand authentication methods.
- Detect unexpected Remote Desktop activity.
- Monitor service account usage.
- Identify unusual authentication patterns.

Changes in normal logon type distribution may indicate unauthorized access or abnormal system behavior.

9.6 Dashboard Benefits

The dashboards created during this project provide several operational benefits:

- Centralized authentication monitoring.
- Faster identification of suspicious activity.
- Improved visibility into user authentication.
- Reduced investigation time.
- Easy visualization of authentication trends.
- Support for proactive threat detection.

These dashboards demonstrate how Splunk Enterprise can transform raw Windows Security Event Logs into actionable security intelligence.

Chapter Summary

The dashboards developed in this project provide an effective way to monitor Windows authentication events using Splunk Enterprise. By visualizing failed logins, successful user activity, privileged account usage, and logon types, the dashboards improve visibility and help SOC analysts detect suspicious authentication behavior more efficiently.

Chapter 10: Alert Configuration and Testing

10.1 Introduction

Monitoring dashboards helps security analysts observe authentication activity, but dashboards require continuous manual observation. To improve security monitoring, Splunk Enterprise allows administrators to configure alerts that automatically notify analysts whenever suspicious conditions are detected.

In this project, an alert was created to identify excessive failed login attempts that could indicate a brute-force attack or password guessing activity.

10.2 Objective of the Alert

The primary objective of the alert was to detect repeated failed authentication attempts generated by Windows Event ID **4625**.

Instead of manually checking dashboards, the alert automatically identifies suspicious authentication activity whenever the predefined threshold is exceeded.

This approach reduces detection time and enables faster incident response.

10.3 Alert Logic

The alert monitors failed Windows authentication events stored in the **main** index.

If the number of failed login attempts exceeds the configured threshold within a specified period, Splunk automatically triggers the alert.

This detection logic is commonly used in Security Operations Centers to identify:

- Brute-force attacks
 - Password guessing attempts
 - Unauthorized authentication activity
 - Abnormal login failures
-

10.4 Alert Workflow

The alert follows the workflow shown below.

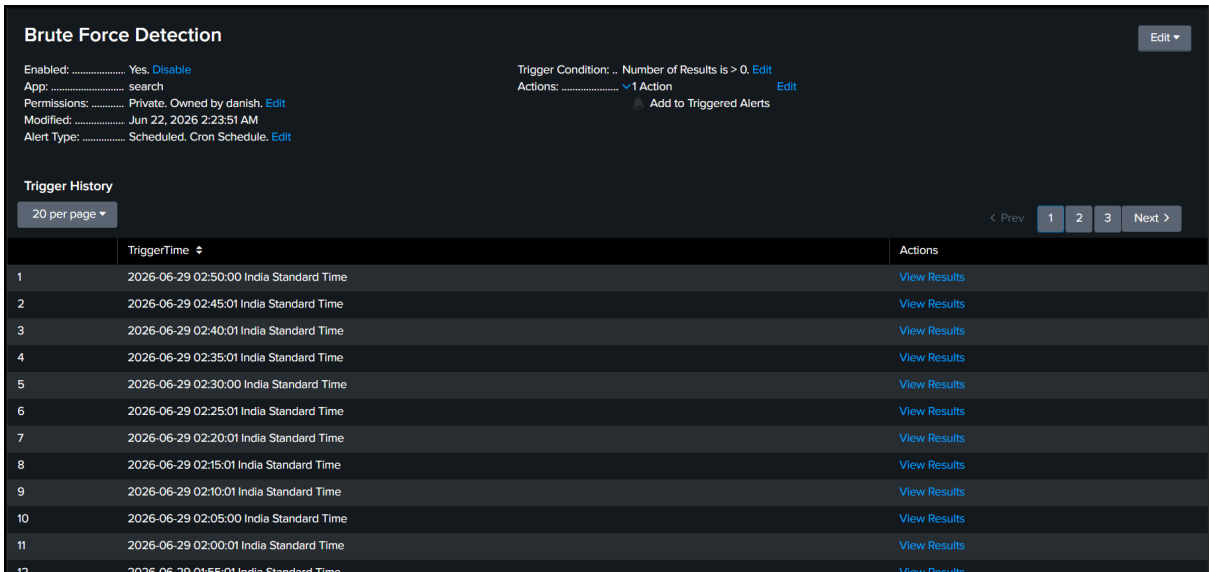
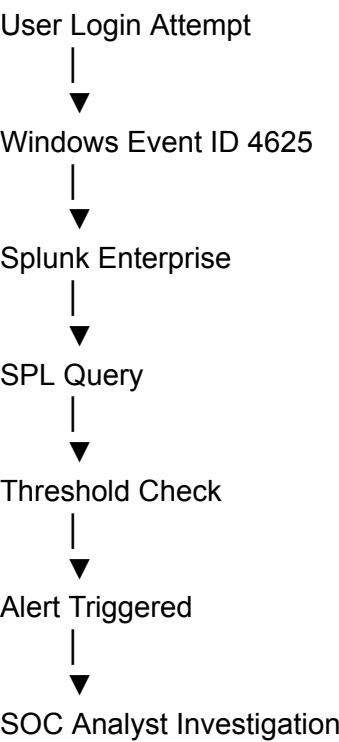


Figure 10.1 – Brute Force Detection Alert

10.5 SOC Investigation Process

Once the alert is triggered, the SOC analyst begins an investigation to determine whether the activity is legitimate or malicious.

The investigation typically includes:

- Reviewing the affected username.
- Identifying the source system.
- Checking the number of failed login attempts.
- Reviewing recent authentication history.
- Looking for successful logins after multiple failures.
- Determining whether the activity matches normal user behavior.

If malicious activity is confirmed, the incident should be escalated according to the organization's security procedures.

10.6 Benefits of Alerting

Configuring alerts provides several advantages during security monitoring.

- Early detection of suspicious authentication activity.
- Reduced manual monitoring.
- Faster incident response.
- Improved visibility into failed login attempts.
- Support for proactive threat detection.

These benefits help SOC analysts identify and respond to authentication-related threats more efficiently.

10.7 Project Outcome

The alert configured in this project successfully demonstrated how Splunk Enterprise can automatically detect suspicious authentication behavior based on Windows Security Event Logs.

By combining automated alerting with dashboards and SPL queries, the project provided a practical example of authentication monitoring within a Security Operations Center (SOC).

Chapter Summary

This chapter explained how Splunk alerts were used to automatically detect suspicious authentication activity. The configured alert demonstrated how repeated failed login attempts could be identified without continuous manual monitoring. Together with the dashboards created in the previous chapter, the alert completed the authentication monitoring workflow implemented in this project.

Chapter 11: Testing and Validation

11.1 Introduction

After configuring Splunk Enterprise, Splunk Universal Forwarder, dashboards, and alerts, the final step was to verify that the entire authentication monitoring solution functioned correctly.

Testing was performed by generating Windows authentication events and confirming that they were successfully collected, indexed, displayed on dashboards, and used to trigger alerts when required.

The purpose of testing was to ensure that every component of the project worked as expected.

11.2 Test Objectives

The testing process was designed to verify the following:

- Windows Security Event Logs were generated successfully.
- Splunk Universal Forwarder forwarded logs to Splunk Enterprise.
- Authentication events were indexed correctly.
- SPL queries returned accurate results.
- Dashboards displayed authentication data correctly.
- Alerts were triggered under the configured conditions.

Successful completion of these tests confirmed that the monitoring solution was functioning correctly.

11.3 Test Scenarios

The following authentication scenarios were tested during the project.

Test Case 1 – Successful Login

Objective

Verify that successful authentication events were collected.

Expected Result

Windows generated Event ID **4624**, and the event appeared in Splunk dashboards.

Status

 Passed

Test Case 2 – Failed Login

Objective

Generate failed authentication attempts.

Expected Result

Windows generated Event ID **4625**, and the failed login appeared in Splunk.

Status

 Passed

Test Case 3 – Dashboard Verification

Objective

Verify that dashboards displayed authentication events correctly.

Expected Result

The following dashboards updated successfully:

- Failed Login Trend
- Top Successful Users
- Privileged Logins
- Logon Type Distribution

Status

 Passed

Test Case 4 – Alert Verification

Objective

Confirm that the configured alert detected repeated failed login attempts.

Expected Result

Splunk generated the configured Brute Force Detection Alert after the defined threshold was reached.

Status

 Passed

11.4 Test Results

The implementation was successfully validated through practical testing.

The following components operated correctly throughout the testing process:

Component	Result
Windows Security Event Logs	 Working
Universal Forwarder	 Working
Splunk Enterprise	 Working
Event Indexing	 Working
SPL Queries	 Working
Dashboards	 Working
Alert Configuration	 Working

The results confirmed that the complete authentication monitoring workflow was operating successfully.

11.5 Project Validation

The project achieved all planned objectives.

Authentication events generated on the Windows machine were successfully forwarded to Splunk Enterprise, indexed into the **main** index, analyzed using SPL queries, visualized through dashboards, and monitored using alerts.

This demonstrated a complete Windows Authentication Monitoring solution similar to those used in Security Operations Centers.

Chapter Summary

Testing and validation confirmed that the Windows Authentication Monitoring project was successfully implemented. Every major component—including log collection, event indexing, dashboard visualization, and alert generation—performed as expected, demonstrating the effectiveness of Splunk Enterprise for monitoring Windows authentication events.

Chapter 12: Troubleshooting

12.1 Introduction

During the implementation of the Windows Authentication Monitoring project, several configuration and connectivity issues were encountered. Troubleshooting these issues was an important part of the learning process because it improved the understanding of Splunk Enterprise, Splunk Universal Forwarder, and Windows Event Log forwarding.

This chapter describes the problems encountered during the project, their root causes, and the steps taken to resolve them.

12.2 Issue 1 – Windows Logs Not Appearing in Splunk

Problem

After configuring Splunk Universal Forwarder, Windows Security Event Logs were not appearing in Splunk Enterprise.

Root Cause

The destination IP address configured in the Universal Forwarder was incorrect. Instead of using the correct Splunk server IP address, an incorrect address was configured, preventing the forwarder from communicating with Splunk Enterprise.

Solution

The following steps were performed:

- Verified the IP address configured in the Universal Forwarder.
- Corrected the destination IP address.
- Restarted the Splunk Forwarder service.
- Verified that the Splunk Enterprise server was reachable.
- Confirmed that Windows Security Event Logs were successfully received.

Result

After correcting the IP address and restarting the service, Windows Security Event Logs were successfully forwarded to Splunk Enterprise.

12.3 Issue 2 – No Results Returned by SPL Queries

Problem

Some SPL searches initially returned no results.

Root Cause

The required authentication events had not yet been generated, resulting in no matching data being available in the selected time range.

Solution

- Generated new login and failed login events.
- Confirmed that events were indexed into the **main** index.
- Expanded the search time range.
- Re-ran the SPL queries.

Result

The queries successfully returned authentication events and dashboards updated correctly.

12.4 Issue 3 – Dashboard Verification

Problem

After creating dashboard panels, some visualizations did not immediately display updated authentication data.

Root Cause

The dashboards depended on newly indexed authentication events. If no recent events were available, the visualizations remained unchanged.

Solution

- Generated additional authentication events.
- Refreshed the dashboard.
- Confirmed that the selected time range included the latest events.

Result

The dashboard panels displayed updated authentication statistics successfully.

12.5 Lessons Learned

Troubleshooting these issues provided valuable practical experience.

Key lessons learned include:

- Always verify IP address configurations before troubleshooting other components.
- Confirm that Splunk Universal Forwarder is successfully forwarding events.
- Verify that authentication events exist before executing SPL queries.
- Check the selected time range when searching for recent events.
- Test each configuration change before moving to the next step.

These lessons improved the overall reliability of the project and strengthened troubleshooting skills required in real SOC environments.

Chapter Summary

Troubleshooting played an important role in the successful implementation of this project. By identifying configuration issues, verifying log forwarding, and validating search results, the Windows Authentication Monitoring solution was successfully completed. These practical experiences reflect common tasks performed by SOC analysts when maintaining SIEM environments.

Chapter 13: Skills Learned

13.1 Introduction

The Windows Authentication Monitoring project provided practical experience in implementing a Security Information and Event Management (SIEM) solution using Splunk Enterprise. Throughout the project, various technical and analytical skills were developed by configuring log collection, creating dashboards, writing SPL queries, and investigating authentication events.

The project also improved understanding of how Security Operations Centers (SOCs) monitor Windows authentication activity and respond to suspicious events.

13.2 Technical Skills Gained

During the implementation of this project, the following technical skills were developed:

- Splunk Enterprise installation and configuration.
 - Splunk Universal Forwarder deployment.
 - Windows Security Event Log monitoring.
 - Search Processing Language (SPL).
 - Log analysis and event searching.
 - Dashboard creation and visualization.
 - Alert configuration.
 - Authentication monitoring.
 - Basic SIEM implementation.
 - Troubleshooting Splunk configuration issues.
-

13.3 SOC Analyst Skills

This project also helped develop practical SOC analyst skills, including:

- Monitoring Windows authentication events.
- Identifying failed login attempts.
- Investigating privileged account activity.
- Detecting possible brute-force attacks.
- Interpreting security dashboards.
- Analyzing authentication trends.
- Responding to security alerts.
- Performing basic incident investigation.

These activities closely reflect the responsibilities of a Security Operations Center (SOC) analyst.

13.4 Project Outcome

By completing this project, the following objectives were successfully achieved:

- Successfully collected Windows Security Event Logs.
- Forwarded authentication events to Splunk Enterprise.
- Indexed and searched authentication logs.
- Created multiple security dashboards.
- Configured an authentication monitoring alert.
- Validated the complete monitoring workflow.
- Documented the implementation process.

The project demonstrated the complete lifecycle of collecting, analyzing, visualizing, and monitoring Windows authentication events using Splunk Enterprise.

Chapter Summary

This project provided practical experience with SIEM implementation, Windows authentication monitoring, and Splunk Enterprise. In addition to technical knowledge, the project strengthened troubleshooting, analytical thinking, and security monitoring skills that are directly applicable to a SOC Analyst role.

Chapter 14: Conclusion

14.1 Project Summary

The **Windows Authentication Monitoring & Threat Detection using Splunk Enterprise** project successfully demonstrated the implementation of a basic Security Information and Event Management (SIEM) solution for monitoring Windows authentication events.

The project involved configuring Splunk Enterprise and Splunk Universal Forwarder to collect Windows Security Event Logs, analyze authentication activities using Search Processing Language (SPL), create interactive dashboards, and configure alerts to detect suspicious login behavior.

By implementing this workflow, the project simulated a real-world Security Operations Center (SOC) environment where authentication events are continuously monitored to improve security visibility and support incident investigation.

14.2 Objectives Achieved

The objectives defined at the beginning of the project were successfully completed.

The following achievements were accomplished:

- Successfully deployed Splunk Enterprise.
- Configured Splunk Universal Forwarder for log collection.
- Collected Windows Security Event Logs.
- Monitored authentication Event IDs (4624, 4625, 4672, and 4740).
- Developed SPL queries for log analysis.
- Created dashboards to visualize authentication activity.
- Configured alerts for failed login detection.
- Tested and validated the monitoring workflow.
- Documented the complete implementation process.

These objectives demonstrate the successful implementation of a Windows Authentication Monitoring solution using Splunk.

14.3 Practical Learning

This project provided valuable hands-on experience with technologies and concepts commonly used by Security Operations Center (SOC) analysts.

Key learning outcomes include:

- Understanding the workflow of a SIEM solution.
- Collecting and analyzing Windows Security Event Logs.
- Writing and optimizing SPL queries.
- Designing dashboards for security monitoring.
- Configuring security alerts.
- Troubleshooting log forwarding issues.
- Performing basic authentication event investigations.

The practical implementation strengthened both technical and analytical skills required for entry-level SOC Analyst roles.

14.4 Future Improvements

Although the project successfully achieved its objectives, several enhancements can be implemented in future versions.

Possible improvements include:

- Integration with Sysmon for advanced endpoint visibility.
- Monitoring additional Windows Event IDs.
- Implementing email notifications for alerts.
- Creating additional dashboards for user behavior analysis.
- Integrating threat intelligence feeds.
- Deploying the solution in a multi-system environment.
- Expanding the project using Microsoft Sentinel or other SIEM platforms.

These improvements would provide broader security monitoring capabilities and simulate larger enterprise environments.

14.5 Final Conclusion

The Windows Authentication Monitoring & Threat Detection project successfully demonstrated how Splunk Enterprise can be used to collect, analyze, and monitor Windows Security Event Logs in a centralized SIEM environment.

The project provided practical experience in authentication monitoring, dashboard creation, alert configuration, and basic security investigation. It also strengthened the understanding of how Security Operations Centers use SIEM platforms to identify suspicious authentication activity and improve organizational security.

This project serves as a strong foundation for further learning in SIEM, incident response, threat hunting, and enterprise security monitoring. The knowledge and practical skills gained through this implementation will support future work in SOC Analyst and Cyber Security roles.

About the Author

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Danish Mansuri is currently pursuing a Master of Computer Applications (MCA) with a specialization in **Cyber Security & Digital Forensics**. He is passionate about Security Operations Center (SOC), Security Information and Event Management (SIEM), threat detection, and incident response.

This project, **Windows Authentication Monitoring & Threat Detection using Splunk Enterprise**, was developed to gain practical experience in implementing a SIEM solution for Windows authentication monitoring. Through this project, hands-on skills were developed in log collection, event analysis, dashboard creation, alert configuration, and troubleshooting using Splunk Enterprise.

Technical Skills

- Splunk Enterprise
- Splunk Universal Forwarder
- Windows Security Event Logs
- Search Processing Language (SPL)
- Dashboard Creation
- Alert Configuration
- SIEM Fundamentals
- Active Directory
- Basic Incident Investigation

Project Repository

GitHub: *(Add your GitHub repository link here)*

Professional Profile

LinkedIn: *(Add your LinkedIn profile link here)*

Document Information

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Thank you for reading this documentation.